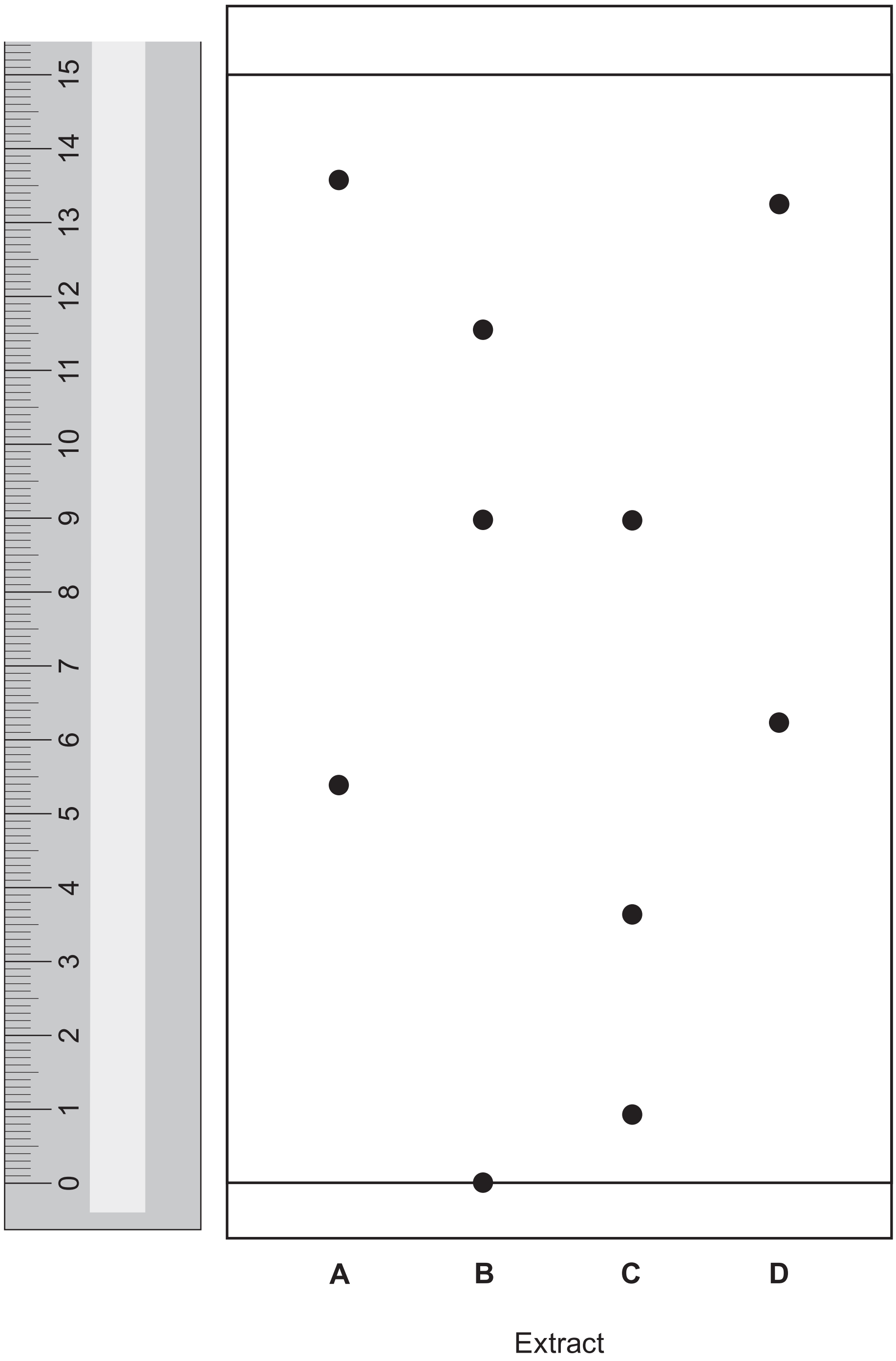
10. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



Examiner only

(a) All of the extracts contain a mixture of pigments with different R_f values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest** R_f value 0.60?

Give your reasoning. [3]

Extract ....

Reasoning ....

....

....

(b) Explain why the pigments travel different distances on the chromatogram. [2]

....

....

....

(c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice. [2]

Extract ....

Explanation ....

....

(d) The chemical formula of the solvent acetone is C_3H_6O. Calculate the percentage by mass of carbon in acetone.

The relative formula mass (M_r) of acetone is 58. [2]

$A_r(C) = 12$

Percentage = ... %

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